HOMEWORK 1

Home advantage

Overall 100pts(including bonus question). Submit both markdown and pdf files

1. Calculate average number of goals for home and away teams per SEASON for that league, use dplyr. (10p)

```
(england_means <- f_data_sm %>%
  filter(COUNTRY == "England") %>%
  group_by(SEASON) %>%
  summarize(MEAN_FTHG = mean(FTHG), MEAN_FTAG = mean(FTAG)))
```

```
## # A tibble: 29 x 3
##    SEASON MEAN_FTHG MEAN_FTAG
##     <dbl>     <dbl>     <dbl>
##  1   1994      1.44      1.15
##  2   1995      1.51      1.08
##  3   1996      1.53      1.07
##  4   1997      1.47      1.08
##  5   1998      1.56      1.12
##  6   1999      1.46      1.07
##  7   2000      1.67      1.12
##  8   2001      1.54      1.07
##  9   2002      1.47      1.17
## 10   2003      1.5       1.13
## # ... with 19 more rows
## # i Use `print(n = ...)` to see more rows
```

2. Calculate average number of goals for home and away teams per SEASON for that league for both top and bottom teams, use dplyr. Is there any difference in between the top and bottom teams in terms of average number of goals ? The function get_top_bottom_teams can be used to get the top and bottom teams (10p)

```
top_england <- get_top_bottom_teams(f_data_sm, 'England')
bottom_england <- get_top_bottom_teams(f_data_sm, 'England', top = F)

(top_england_mean_goals <- top_england %>%
    group_by(SEASON) %>%
      summarize(MEAN_FTHG = mean(FTHG), MEAN_FTAG = mean(FTAG)))
```

```
## # A tibble: 29 x 3
##    SEASON MEAN_FTHG MEAN_FTAG
##     <dbl>     <dbl>     <dbl>
```

```
##  1   1994      1.52     1.05
##  2   1995      1.5      1.16
##  3   1996      1.49     0.933
##  4   1997      1.56     1.31
##  5   1998      1.7      1.06
##  6   1999      1.37     1.14
##  7   2000      1.57     1.06
##  8   2001      1.68     1.07
##  9   2002      1.68     1.32
## 10   2003      1.73     1.29
## # ... with 19 more rows
## # i Use `print(n = ...)` to see more rows
```

```
(bottom_england_mean_goals <- bottom_england %>%
    group_by(SEASON) %>%
      summarize(MEAN_FTHG = mean(FTHG), MEAN_FTAG = mean(FTAG)))
```

```
## # A tibble: 29 x 3
##    SEASON MEAN_FTHG MEAN_FTAG
##     <dbl>     <dbl>     <dbl>
##  1   1994      1.40     1.18
##  2   1995      1.35     0.985
##  3   1996      1.39     1.11
##  4   1997      1.34     1.01
##  5   1998      1.33     1.15
##  6   1999      1.36     0.973
##  7   2000      1.82     1.05
##  8   2001      1.37     1.06
##  9   2002      1.23     1.02
## 10   2003      1.47     1.12
## # ... with 19 more rows
## # i Use `print(n = ...)` to see more rows
```

3.Now calculate the same statistics for the games where the hometeam was in bottom of the table and the away team was in the top of the table. Is the home advantage still a strong factor ? (10p)

```
england_top_vs_bottom <- get_top_vs_bottom_teams(f_data_sm, 'England')

(top_vs_bottom_england_mean_goals <- england_top_vs_bottom %>%
    group_by(SEASON) %>%
      summarize(MEAN_FTHG = mean(FTHG), MEAN_FTAG = mean(FTAG)))
```

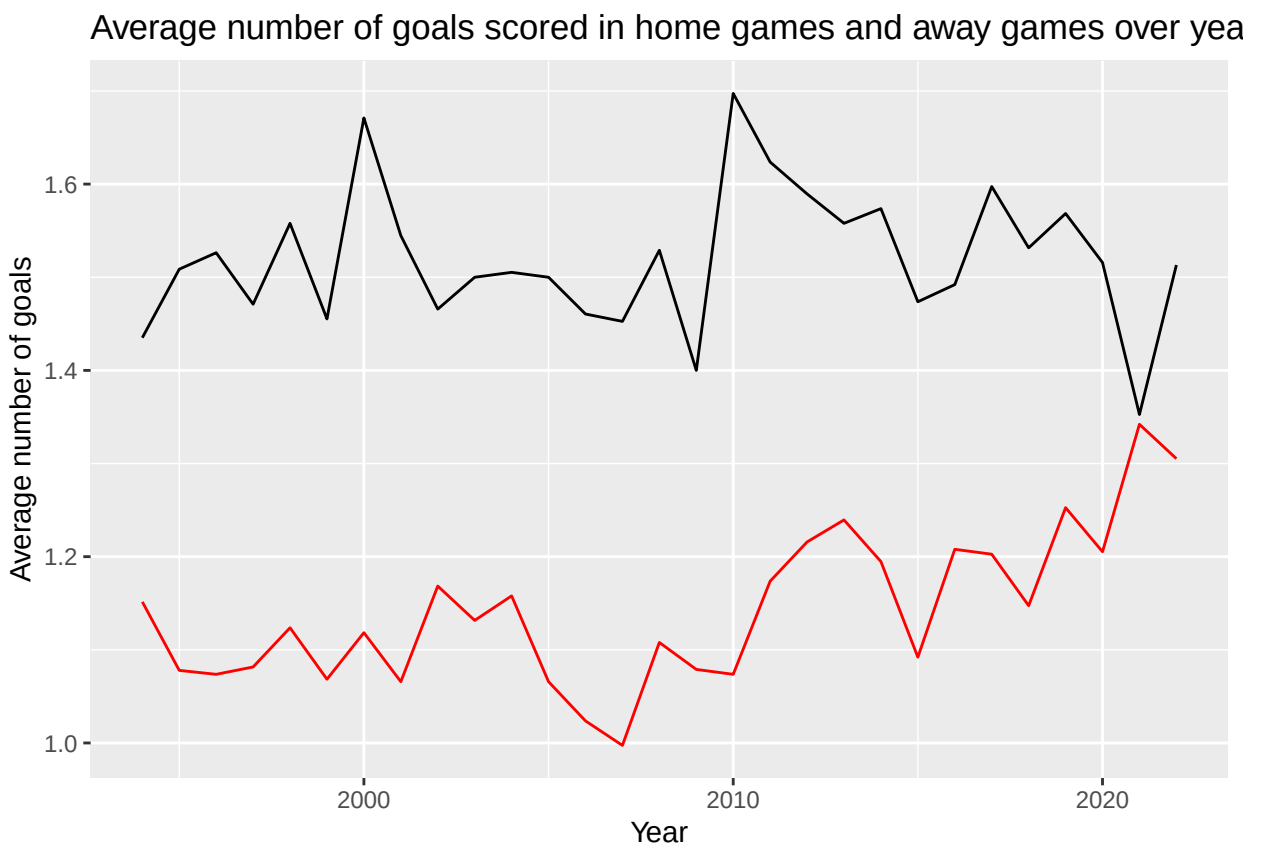
```
## # A tibble: 29 x 3
##    SEASON MEAN_FTHG MEAN_FTAG
##     <dbl>     <dbl>     <dbl>
##  1   1994      1.05      1.55
##  2   1995      1.27      1.47
##  3   1996      1.15      1.49
##  4   1997      1.17      1.26
##  5   1998      1.14      1.46
##  6   1999      1.04      1.33
##  7   2000      1.27      1.56
##  8   2001      1.08      1.34
```

```
##  9   2002      1.09      1.55
## 10   2003      1.15      1.49
## # ... with 19 more rows
## # i Use 'print(n = ...)' to see more rows
```

4. Construct a plot using ggplot to show how this number is changing over time. Use the means from the first exercise.

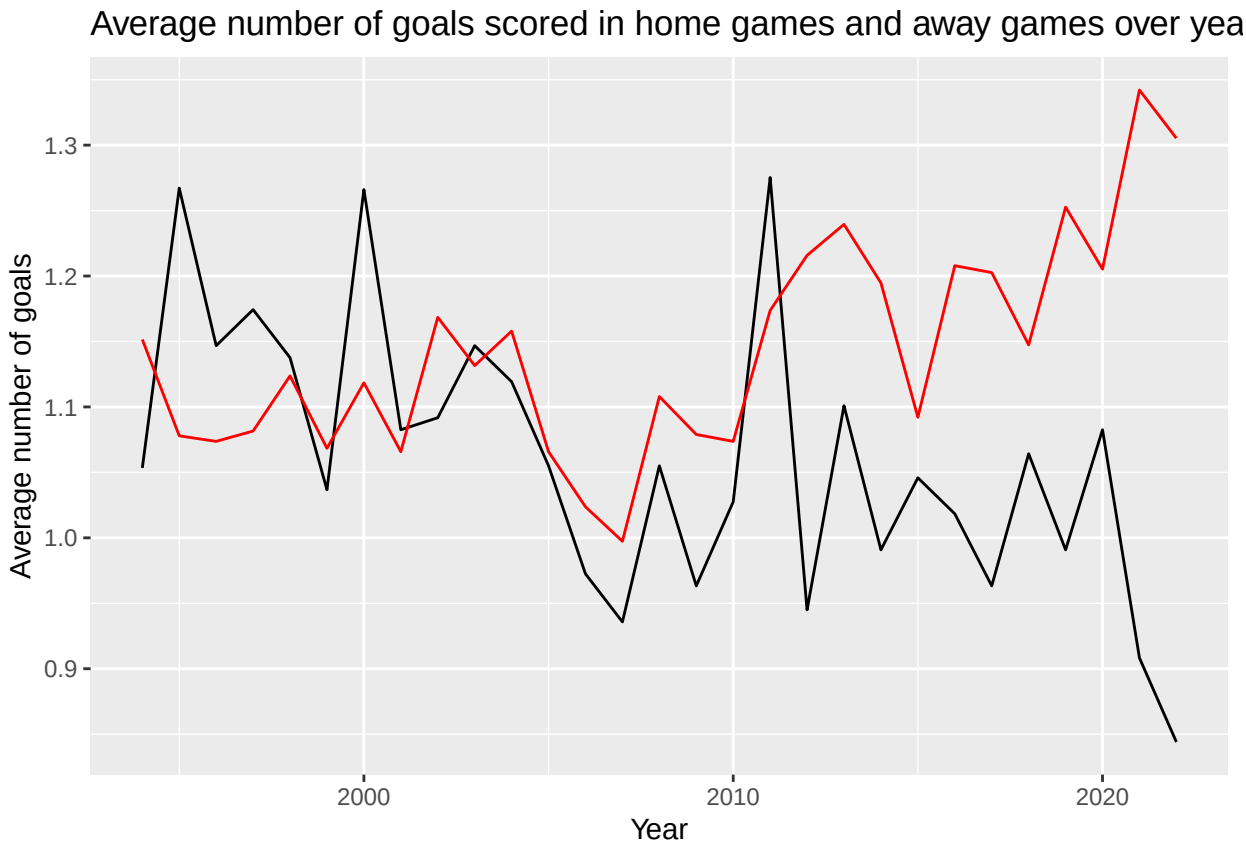
Note: you need to have SEASON on x-axis. Show average Home goals and Away goals on the same plot. Be sure that your plot has appropriate axis names and title (10p)

```
ggplot(england_means, aes(x = SEASON, y = MEAN_FTHG)) +
  geom_line() +
  geom_line(data = england_means, aes(x = SEASON, y = MEAN_FTAG), color = 'red') +
  labs(title = "Average number of goals scored in home games and away games over years") +
  xlab("Year") +
  ylab("Average number of goals")
```



5. Now use the same plot on the games where the away team was from top of the table (10p)

```
ggplot(top_vs_bottom_england_mean_goals, aes(x = SEASON, y = MEAN_FTHG)) +
  geom_line() +
  geom_line(data = england_means, aes(x = SEASON, y = MEAN_FTAG), color = 'red') +
  labs(title = "Average number of goals scored in home games and away games over years") +
  xlab("Year") +
  ylab("Average number of goals")
```



6. Interpret the plots (10p)

7. Think of your own approach on how will you measure home team advantage given the data you have. Calculate Home team advantage for your league and for all other leagues over time (10p)

```
(england_means <- england_means %>%
  mutate(HADV = (MEAN_FTHG / MEAN_FTAG) * (MEAN_FTHG - MEAN_FTAG)))
```

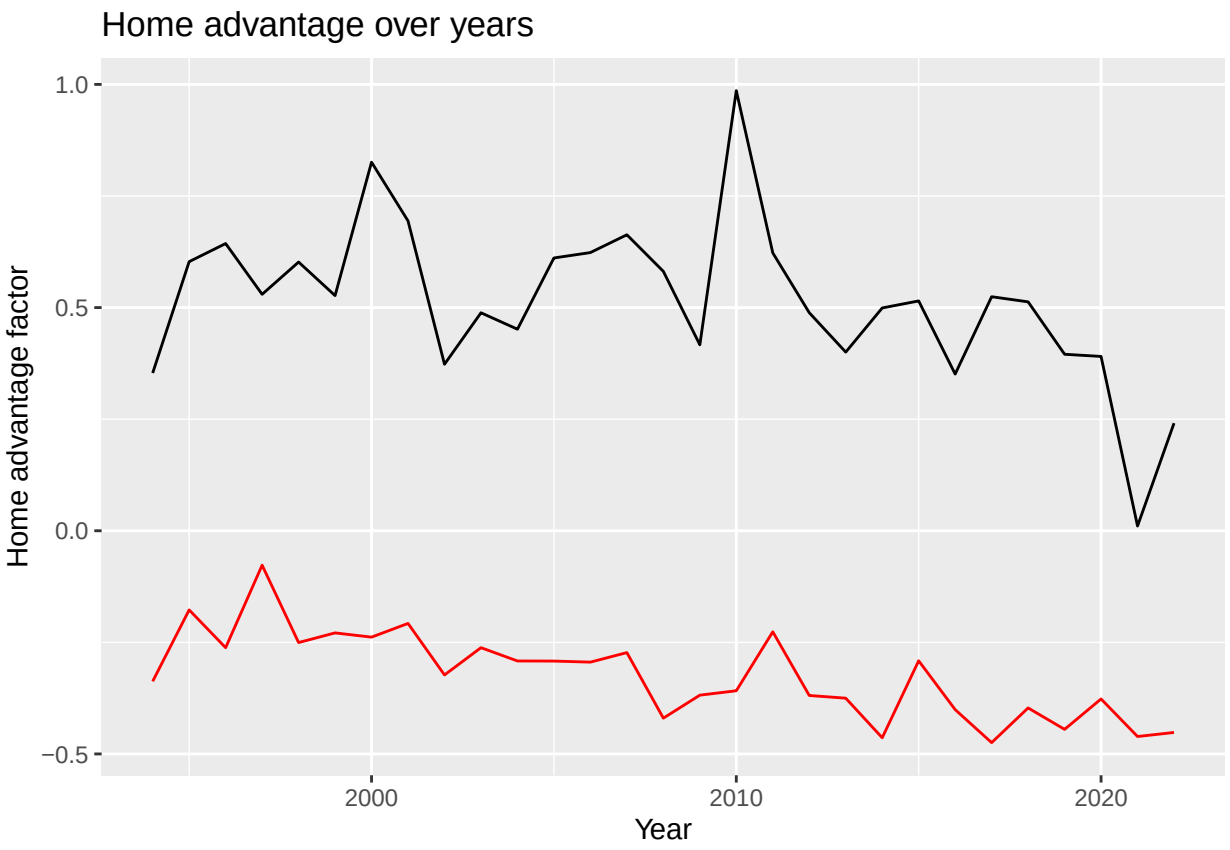
```
## # A tibble: 29 x 4
##    SEASON MEAN_FTHG MEAN_FTAG  HADV
##     <dbl>     <dbl>     <dbl> <dbl>
##  1   1994      1.44      1.15 0.353
##  2   1995      1.51      1.08 0.603
##  3   1996      1.53      1.07 0.643
##  4   1997      1.47      1.08 0.530
##  5   1998      1.56      1.12 0.602
##  6   1999      1.46      1.07 0.527
##  7   2000      1.67      1.12 0.826
##  8   2001      1.54      1.07 0.694
##  9   2002      1.47      1.17 0.373
## 10   2003      1.5       1.13 0.488
## # ... with 19 more rows
## # i Use 'print(n = ...)' to see more rows
```

8. Calculate the same ratio for the games where the away team was from the top of the table and the home team was from the bottom of the table. (10p)

```
top_vs_bottom_england_mean_goals <- top_vs_bottom_england_mean_goals %>%
  mutate(HADV = (MEAN_FTHG / MEAN_FTAG) * (MEAN_FTHG - MEAN_FTAG))
```

9. Plot the results for all games and the games where the away team was from the top of the table in one plot using ggplot2. (10p)

```
ggplot(england_means, aes(x = SEASON, y = HADV)) +
  geom_line() +
  geom_line(data = top_vs_bottom_england_mean_goals, aes(X= SEASON, y = HADV), color = 'red') +
  labs(title = "Home advantage over years") +
  xlab("Year") +
  ylab("Home advantage factor")
```



Predictions

1. With your chosen league, pick an upcoming game and calculate the betting odds for that game. Use distribution approach. You can compare the result with the actual betting odds from this website, https://www.bet365.com/#/AS/B1/. (10p)

```
england_home_2022 <- f_data_sm %>%
  filter(SEASON == "2022", COUNTRY == "England") %>%
  group_by(HOMETEAM) %>%
  summarise(MEAN_FTHG = mean(FTHG))
```

```r
england_away_2022 <- f_data_sm %>%
  filter(SEASON == "2022", COUNTRY == "England") %>%
  group_by(AWAYTEAM) %>%
  summarise(MEAN_FTAG = mean(FTAG))

# Arsenal vs Totenham
arsenal_fthg_mean <- england_home_2022 %>%
  filter(HOMETEAM == "Arsenal") %>%
  pull(MEAN_FTHG)

arsenal_ftag_mean <- england_away_2022 %>%
  filter(AWAYTEAM == "Arsenal") %>%
  pull(MEAN_FTAG)

rs <- rskellam(10000,mu1 = arsenal_fthg_mean, mu2 = arsenal_ftag_mean)
HTW <- sum(rs>0)/10000
D <- sum(rs == 0)/10000
ATW <- sum(rs<0)/10000

(coeff <- data.frame(HW = 1/HTW, D = 1/D, ATW = 1/ATW))
```

```
##         HW        D      ATW
## 1 2.085506 4.249894 3.506311
```

```r
# Actual coeff
# 2.05, 3.60, 3.25
```

Power Ratings

1.Try predicting home results with power ratings in other sports, Does it work as well as in football ? Try to interpret the actual results. The data for nba, nfl is available in SportsAnalytics270 (Bonus question, 10 points)

```r
library(dplyr)
library(zoo)

data("nba2009_2021")
nba_2021 <- nba2009_2021 %>%
  filter(SEASON_ID == '2021') %>%
  arrange(GAME_DATE) %>%
  select("SEASON_ID","GAME_DATE","home.TEAM_NAME","away.TEAM_NAME","home.PTS",
         "away.PTS", "home.WL") %>%
  mutate(ID = 1:n())

nba_2021_long <-  nba_2021 %>% reshape2::melt(
                    id.vars = c("ID", "GAME_DATE", "SEASON_ID",
                                "home.TEAM_NAME", "away.TEAM_NAME","home.WL"),
                    measure.vars = c("home.PTS","away.PTS"),
                    variable.name="H_A") %>%
                    arrange(ID)

nba_2021_long <- nba_2021_long %>%
```

```
  mutate(TEAM1 = ifelse(H_A == 'home.PTS', home.TEAM_NAME, away.TEAM_NAME),
         TEAM2 = ifelse(H_A == 'home.PTS', away.TEAM_NAME, home.TEAM_NAME)) %>%
  select(ID, SEASON_ID, GAME_DATE, TEAM1, TEAM2, H_A, value, home.WL)


nba_2021_long <- nba_2021_long %>%
  group_by(SEASON_ID, TEAM1) %>%
  mutate(CUM_SC = dplyr::lag(rollapply(value %/% 2.5 ,6,sum, fill=NA, align='right'),1)) %>%
  ungroup() %>%
  group_by(SEASON_ID, TEAM2) %>%
  mutate(CUM_C = lag(rollapply(value %/% 2.5 ,6,sum, fill=NA, align='right'),1)) %>%
  filter(CUM_SC != "NA", CUM_C != "NA") %>%
  as.data.frame()

nba_2021_wide = reshape(nba_2021_long, direction="wide",
  timevar = "H_A", idvar = "ID",
  v.names=c("value", "CUM_SC", "CUM_C"))

nba_2021_wide = nba_2021_wide %>%
  mutate(POWERHOME = CUM_SC.home.PTS - CUM_C.home.PTS,
         POWERAWAY = CUM_SC.away.PTS - CUM_C.away.PTS,
         POWERDIFF = POWERHOME - POWERAWAY,
         RESULT = ifelse(home.WL == 'W', 'HW', 'AW')) %>%
  select(SEASON_ID, GAME_DATE, TEAM1, TEAM2, value.home.PTS, value.away.PTS,
         POWERHOME, POWERAWAY, POWERDIFF, RESULT) %>%
  filter(!is.na(POWERDIFF))

freqs = dcast(nba_2021_wide, POWERDIFF~RESULT)
head(freqs)
```

```
##   POWERDIFF AW HW
## 1      -122  0  1
## 2       -90  0  1
## 3       -86  0  2
## 4       -83  1  0
## 5       -78  1  1
## 6       -77  0  1
```

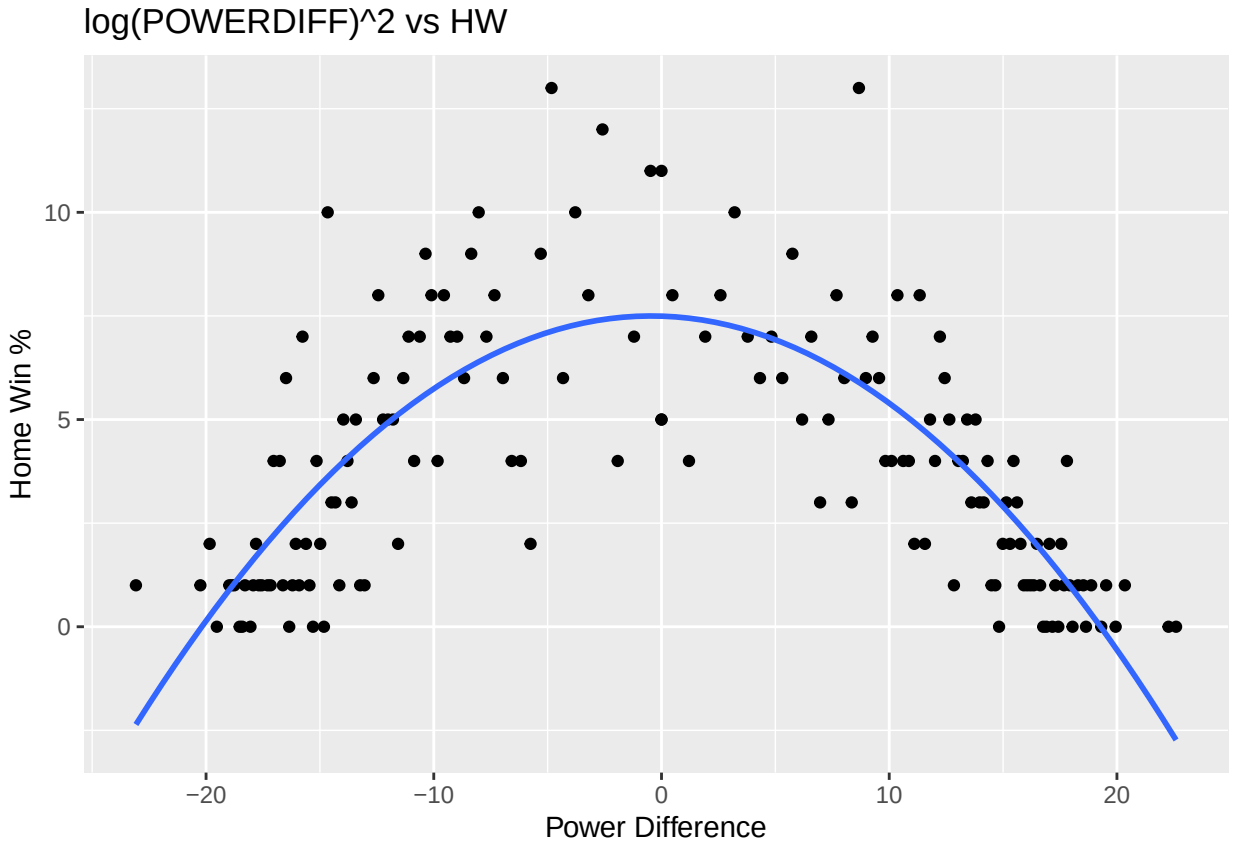
```
freqs[freqs$POWERDIFF==0,]
```

```
##    POWERDIFF AW HW
## 78         0  3  5
```

```
freqs$POWERDIFF_LOG = ifelse(freqs$POWERDIFF > 0, log(freqs$POWERDIFF),
                             ifelse(freqs$POWERDIFF == 0, 0, -log(-freqs$POWERDIFF)))

freqs$POWERDIFF_LOG_SQUARED = ifelse(freqs$POWERDIFF_LOG > 0,
                                     freqs$POWERDIFF_LOG^2, - freqs$POWERDIFF_LOG^2)


ggplot(freqs, aes(x=POWERDIFF_LOG_SQUARED, y=HW))+geom_point()+
geom_smooth(method="lm",formula = y ~ poly(x, 2), se=F)+
labs(x="Power Difference", y="Home Win %", title = "log(POWERDIFF)^2 vs HW")
```




```
modelH = lm(HW~poly(POWERDIFF_LOG,2), data=freqs)
summary(modelH)
```

```
##
## Call:
## lm(formula = HW ~ poly(POWERDIFF_LOG, 2), data = freqs)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.0484 -1.2128 -0.2124  1.2797  7.6382
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                3.9114     0.1664  23.510   <2e-16 ***
## poly(POWERDIFF_LOG, 2)1   -4.0431     2.0912  -1.933    0.055 .
## poly(POWERDIFF_LOG, 2)2 -29.5731     2.0912 -14.142   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.091 on 155 degrees of freedom
## Multiple R-squared:  0.5679, Adjusted R-squared:  0.5623
## F-statistic: 101.9 on 2 and 155 DF,  p-value: < 2.2e-16
```